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Attachments to the Public Switched Telephone Network (PSTN)

Category II attachment requirements for 1200 bits per second half duplex and 1200/75 bits per second asymmetrical duplex modems standardized for use on the PSTN

(The text of this ETS may be utilized, wholly or in part, for the establishment of NET 24)

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Contents

Foreword 5

1 Scope 7

2 Normative references 7

3 Definitions and abbreviations 7
3.1 Definitions 7
3.2 Abbreviations 8

4 General requirements 9
4.1 References to other ETSs 9
4.2 Information to be provided by the applicant 9
4.2.1 Information required for testing purposes 9
4.2.2 Instructions for use 9

5 Functional requirements specific to Category II modems 9
5.1 General requirements 9
5.2 Modes of operation/use 10
5.3 Line signals 10
5.3.1 Carrier frequencies 10
5.3.2 Spectral power 10
5.4 Line signalling rates 11
5.5 Channel allocation 11
5.6 Response times 11
5.6.1 Response times of the forward channel and backward channel received line signal detectors 11
5.6.2 Response times of the forward channel ready for sending indicator 12
5.6.2.1 For modems with DTE control 12
5.6.2.2 For modems with control derived from the backward channel 12
5.6.2.3 For modems which transmit carrier constantly when in the on-line state 13
5.6.3 Response times of the backward channel ready for sending indicator 13
5.6.3.1 For modems with DTE control 13
5.6.3.2 For modems with control derived from the forward channel 13
5.7 Synchronous mode of use 14
5.8 Thresholds of received line signal detectors 14
5.9 Auto-calling and auto-answering procedures 14
5.9.1 Auto-calling - calling tone 14
5.9.2 Auto-calling- recognition of answering tone 14
5.9.3 Auto-answering - answer tone 14
5.10 Provision of test looping facilities 15
5.11 Performance 15
5.11.1 Distortion 15
5.11.2 Receiver performance 15

Annex A (Normative): Testing methods 16

A.1 General testing conditions 16

A.2 Test for subclause 5.3.1 (Carrier frequencies) 16

A.3 Test for subclause 5.5 (Channel allocation)..... 16

A.4 Test for subclause 5.6 (Response times) 16
A.4.1 Test for subclause 5.6.1 17
A.4.2 Test for subclause 5.6.2 (Data channel Ready For Sending (RFS) response times)..... 17
A.4.2.1 Test for subclause 5.6.2.1..... 17
A.4.2.2 Test for subclause 5.6.2.2..... 17
A.4.2.3 Test for subclause 5.6.2.3..... 17
A.4.3 Test for subclause 5.6.3 (Backward channel RFS response times) 18
A.4.3.1 Test for subclause 5.6.3.1..... 18
A.4.3.2 Test for subclause 5.6.3.2..... 18

A.5 Test for subclause 5.7 (Synchronous mode of use)..... 18

A.6 Test for subclause 5.11.1 (Distortion) 18
A.6.1 Test for subclause 5.11.1 (a) 19
A.6.2 Test for subclause 5.11.1 (b) 19
A.6.3 Test for subclause 5.11.1 (c) 19

Annex B (Informative): Proforma declaration of modes of operation/use..... 20

History 22

Foreword

This European Telecommunication Standard (ETS) was produced by the Terminal Equipment (TE) Technical Committee of the European Telecommunications Standards Institute (ETSI) and adopted in November 1990. The text of this draft ETS may be utilized, wholly or in part, for the establishment of NET 24.

This ETS contains the technical requirements for approval to Category II of 1200 bits per second half duplex and 1200/75 bits per second asymmetrical duplex modems standardized for use on the Public Switched Telephone Network (PSTN). Approval and testing to Category II is applicable only at the request of the applicant. These requirements are based upon, and do not conflict with, CCITT Recommendation V.23 [4] except in the case of subclause 5.3.1 where the frequency tolerances specified are tighter than those given.

Additionally, requirements are included relating to end-to-end inter-operability over PSTN connections. Such requirements are in excess of the CCITT Recommendations. A modem which complies with CCITT Recommendation V.23 [4] should always meet the requirements of this ETS which relate to parameters specified in that CCITT Recommendation.

Clause 4 of this ETS references the requirements common to both Category I and II modems, which are contained in ETS 300 114 [2], Clause 4.

Clause 5 of this ETS contains Category II requirements specific to 1200 bits per second half duplex and 1200/75 asymmetrical duplex modems. In the case of certain functions common to a number of different types of modem (e.g. Auto-answering sequence) reference is made to Clause 5 of ETS 300 114 [2] which contains the relevant requirements.

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1 Scope

This European Telecommunication Standard (ETS) specifies the technical characteristics to be met by a modem seeking Category II approval for

- half duplex operation over the PSTN at 1200 bits per second; and/or

- asymmetrical duplex operation over the PSTN at 1200/75 bits per second.

The modulation schemes specified are those described in CCITT Recommendation V.23 [4].

The term "modem" in the context of this ETS includes all physical implementation practices for a voice band modem, which are galvanically connected to the PSTN.

This ETS specifies four modes of operation/use (see subclause 5.2).

This ETS also contains descriptions of the tests to be performed in order to confirm compliance with the functional requirements contained herein. A general description of the test conditions and test requirements is given in Annex A (Normative).

NOTE: To aid reading, the term "forward channel" is sometimes used to describe the channel capable of transmitting up to 1200 bit/s, and the term "backward channel" is similarly used to describe the channel capable of transmitting up to 75 bit/s.

2 Normative references

This ETS incorporates, by dated or undated reference, provisions from other publications. These normative references are cited at the appropriate places in the text and the publications are listed hereafter. For dated references, subsequent amendments to, or revisions of, any of these publications apply to this ETS only when incorporated in it by amendment or revision. For undated references the latest edition of the publication referred to applies.

[1] Draft prETS 300 001, "Attachments to Public Switched Telephone Network (PSTN). General technical requirements for equipment connected to an analogue subscriber interface in the PSTN (candidate NET4)"

[2] ETS 300 114 (1990), "Attachments to the Public Switched Telephone Network (PSTN), Basic attachment requirements for modems standardized for use on the PSTN"

[3] CCITT Recommendation S.33 (1984), "Standardization of an international text for the measurement of the margin of start-stop machines using International Alphabet No 5"

[4] CCITT Recommendation V.23 (1984), "600/1200-Baud modem standardized for use in the General Switched Telephone Network"

3 Definitions and abbreviations

For the purposes of this standard the definitions and abbreviations contained in ETS 300 114 [2] shall apply, together with the following.

3.1 Definitions

Data Terminal Equipment (DTE): the expression "DTE" is used to define the origin and destination of signals present at the digital interface of a modem. This expression does not require that a "commercial data terminal" be present to receive or generate such signals; a tester or any other suitable device may monitor or generate such signals.

Degree of start-stop distortion: in start-stop transmission the ratio of the maximum measured difference, irrespective of sign, between the actual and theoretical intervals separating any significant instant from the significant instant of the start element immediately preceding it, to the unit interval. The highest absolute value of degrees of individual distortion of the significant instants of a start-stop signal is reached within a specific time interval. The degree of distortion of start-stop modulation, restitution or signal shall be expressed as a percentage. The result of measurement shall be completed by an indication of the period of the observation. The start-stop distortion shall be considered positive when the significant instant occurs after the ideal instant and conversely, negative when it occurs before.

Degree of synchronous start-stop distortion: the degree of start-stop distortion determined when the assumed unit interval is that appropriate to the actual modulation rate. The degree of synchronous start-stop distortion shall be measured by adjusting the scanning rate of the distortion measuring set. The start-stop distortion shall be considered positive when the significant instant occurs after the ideal instant and conversely, negative when it occurs before. For the determination of the actual mean modulation rate, account shall only be taken of those significant instants of modulation (or restitution) that correspond to a change on the same sense as that occurring at the beginning of the start element.

Modem: a functional unit that modulates and demodulates signals in order to enable digital data to be transmitted over analogue transmission facilities.

Modem used for reference: a modem used for some of the tests specified herein or in another modem specific ETS. A modem used for reference may, at the discretion of the applicant, be provided by the testing authority or by himself. It shall be designed:

- to meet the requirements of the same Recommendation(s) of the CCITT as the modem under test, to the extent necessary for performing the tests;

- to provide the functionalities for a modem used for reference that are specified in the relevant testing clauses; and

- to provide an interface which is accessible and of a type suitable for use in the tests (e.g. CCITT Recommendation V.24).

Where the applicant has provided the modem used for reference and the test fails, the testing authority may not be in a position to determine the precise reason for failure.

Modes of operation: in the context of this ETS, modes of operation are defined as modes specified in this modem-specific ETS that have an influence upon line signals present at the PSTN interface.

Modes of use: in the context of this ETS, modes of use are defined as modes specified in this modem-specific ETS that have an influence upon conditions present at a digital interface. Examples include a "conventional" CCITT Recommendation V.24 interface or a PC bus interface in the case of an integral modem.

On-line state: an electrical condition into which, when connected to the network, a modem is placed such that it draws enough current to be capable of activating the exchange.

NOTE: Usually, a modem in the on-line state is potentially capable of sending or receiving speech-band information to or from the network.

3.2 Abbreviations

CCITT	Comité Consultatif International Télégraphique et Téléphonique
Cct	Circuit
CEPT	Conférence Européenne des Administrations des Postes et des Télécommunications
DCE	Data Circuit-Terminating Equipment
DTE	Data Terminal Equipment

ISO	Organization for International Standardization (French Terminology)
PSTN	Public Switched Telephone Network
RFS	Ready For Sending

4 General requirements

4.1 References to other ETSs

The modem shall comply with ETS 300 114 [2], Clause 4.

NOTE: ETS 300 114 in turn refers to Draft prETS 300 001 [1] for the majority, if not all, of its requirements.

4.2 Information to be provided by the applicant

4.2.1 Information required for testing purposes

The applicant shall declare for which of the modes of operation/use identified in this ETS approval to Category II is sought.

Compliance will be considered to have been accomplished by provision of the relevant information.

NOTE : This may be carried out by completion of forms such as those provided in Annex B (Informative).

4.2.2 Instructions for use

Instructions for use shall be made available with the apparatus. The instructions for use shall include:

a) the apparatus or types of apparatus to which the instructions apply;

b) any information specifically indicated in this ETS for inclusion in the "Instructions for use"; and

c) any national restrictions on the use of the apparatus.

For the purpose of compliance with this standard, any additional information shall be disregarded unless it is subject of another ETS.

Compliance shall be checked by inspection.

5 Functional requirements specific to Category II modems

5.1 General requirements

To comply with the requirements of this ETS, it shall be necessary that the modem provide

- half-duplex mode with switched carrier or asymmetrical duplex mode of operation/use with continuous carriers, as specified in CCITT Recommendation V.23 [4],

- frequency shift binary modulation with synchronous or asynchronous mode of use,

- signalling rates of up to 1200 bit/s in the forward channel and, optionally, of up to 75 bit/s in the backward channel.

NOTE 1: In the following, references are made to interchange circuits between the modem and the DTE, as defined in CCITT Recommendation V.24. DTEs using certain customised modems will generally not provide an interface with such interchange circuits. For these cases the references to V.24 type interchange circuits indicate equivalent functionality of a DTE and a modem.

NOTE 2: With asymmetrical duplex modems providing a CCITT Recommendation V.24 interface, it is common practice to provide the interchange circuit for the 75 bit/s channel on the relevant pin for the 1200 bit/s channel.

5.2 Modes of operation/use

It shall be possible to configure the modem to operate in at least one of the following modes of operation/use:

a) 1200/1200 bit/s half duplex asynchronous

b) 1200/75 bit/s asymmetrical duplex asynchronous

c) 75/1200 bit/s asymmetrical duplex asynchronous

d) 1200/1200 bit/s half duplex synchronous

It is the applicant's responsibility to specify for which of these modes of operation/use approval for Category II is sought.

NOTE 1: In the above representation, the first number indicates the rate at which data is received, and the second number indicates the rate at which data is transmitted.

NOTE 2: In this ETS, it is not possible to make a clear distinction between mode of operation and mode of use, and therefore no attempt has been made to separate the above list into these categories.

5.3 Line signals

The channels defined are within the telephone bandwidth.

NOTE: The allocation of channels is described in subclause 5.5.

5.3.1 Carrier frequencies

The nominal characteristic frequencies are listed in Table 1:

Table 1: Frequency allocation

	Binary 0 (Fa)	Binary 1 (Fz)
1200 bit/s chl.	2100 Hz	1300 Hz
75 bit/s channel	450 Hz	390 Hz

For modes a), c) and d), as defined in subclause 5.2, the characteristic frequencies Fa and Fz as measured at the line terminals of the modem shall not deviate by more than ± 3 Hz for the 1200 bit/s channel; and for mode b) ± 2 Hz for the 75 bit/s channel (if provided in the modem) from those given in Table 1.

NOTE : These frequency tolerance are tighter than those specified by CCITT Recommendation V.23 [4].

Compliance shall be checked by the method described in Annex A, Clause A.2

5.3.2 Spectral power

The national network dependent spectral power limits are given in ETS 300 114 [2], Clause 4.

NOTE : This refers in turn to Draft prETS 300 001 [1].

5.4 Line signalling rates

Where the modem provides the relevant mode(s) of operation, it shall be capable of transmitting and/or receiving data at line signalling rates of:

- up to 1200 bit/s in modes (a), (b) and (c);

- up to 75 bit/s in modes (b) and (c); and

- 1200 bit/s in mode (d).

Compliance shall be checked during the tests given in subclause 5.11.

5.5 Channel allocation

For modems capable of alternating between modes b) and c) (asymmetrical duplex modems) whilst in the on-line state, two methods of controlling transmission in the 75 bit/s channel are recognized for asymmetrical duplex modems, either:

a) the modem shall enable transmission on the 1200 bit/s channel when the DTE indicates that it wishes to transmit on the 1200 bit/s channel (equivalent: turning ON Cct 105); and the modem shall enable transmission on the 75 bit/s channel when the DTE indicates that it no longer wishes to transmit on the 1200 bit/s channel (equivalent: turning OFF Cct 105);

Compliance shall be checked by the method described in Annex A, Clause A.3.

or,

b) the control of transmission on the 75 bit/s channel shall be on a separate circuit (equivalent: Cct 120) to that for the 1200 bit/s channel, in which case the response times of the ready for sending indicators shall comply with subclauses 5.6.2.2 and 5.6.3.2.

Compliance is checked during performance of the test associated with subclauses 5.6.2.2 and 5.6.3.2.

5.6 Response times

5.6.1 Response times of the forward channel and backward channel received line signal detectors

The response times of the 1200 bit/s channel and the 75 bit/s channel received line signal detectors are the times that elapse between :

a) For the OFF to ON transition: the application of a tone corresponding to binary 1 at a level higher than -43 dBm to the line terminals of the modem and an indication being given to the DTE that a carrier signal has been detected (equivalent: turning ON Cct 109).

b) For the ON to OFF transition: removal of a tone corresponding to binary 1 from, or the level falling instantaneously to below -48 dBm, at the line terminals of the modem and an indication being given to the DTE that a carrier signal is no longer being detected (equivalent: turning OFF Cct 109).

The response times of the detector shall be within the limits stated in Table 2.

Compliance shall be checked by the method described in Annex A subclause A.4.1.

Table 2: Received line signal detector response times

Data Channel Signal Detector	OFF to ON	300 ms to 2000 ms 10 ms to 20 ms (see Note 1 below)
	ON TO OFF	5 ms to 15 ms
Backward Chl. Signal Detector	OFF to ON	< 80 ms
	ON TO OFF	15 ms to 120 ms (see Note 2 below)
NOTE 1: The longer range of response times is only to be used during call establishment in association with automatic calling and answering.		
NOTE 2: CCITT Recommendation V.23 [4] suggests a value of 15 ms to 80 ms.		

5.6.2 Response times of the forward channel ready for sending indicator

The applicant shall state whether the ready for sending delay is either a) or b) or both of Table 3, and that information shall be included in the instruction for use.

The response times of the 1200 bit/s channel ready for sending indicator are those given in subclauses 5.6.2.1 to 5.6.2.3.

5.6.2.1 For modems with DTE control

The response times of the 1200 bit/s channel ready for sending indicator are the times that elapse between:

a) for the OFF to ON transition: a DTE indicating that it wishes to transmit in the 1200 bit/s channel (equivalent: turning ON Cct 105) and an indication being given to the DTE that the 1200 bit/s channel is ready (equivalent: turning ON Cct 106), for modes a), c) and d);

b) for the ON to OFF transition: a DTE indicating that it no longer wishes to transmit in the 1200 bit/s channel (equivalent: turning OFF Cct 105) and an indication being given to the DTE that the 1200 bit/s channel is no longer ready (equivalent: turning OFF Cct 106), for modes a), c) and d).

Compliance shall be checked by the method described in Annex A subclause A.4.2.1

5.6.2.2 For modems with control derived from the backward channel

For a modem in which control of the transmitted carrier in the 1200 bit/s channel is controlled by the reception of signals from the remote modem in the 75 bit/s channel, the response times of the 1200 bit/s channel ready for sending indicator are the times that elapse between:

a) for the OFF to ON transition: an indication to the DTE that a carrier signal has been received in the 75 bit/s channel (equivalent: turning ON Cct 122) and an indication being given to the DTE that the 1200 bit/s channel is ready (equivalent: turning ON Cct 106);

b) for the ON to OFF transition; an indication to the DTE that a carrier signal is no longer being received in the 75 bit/s channel (equivalent: turning OFF Cct 122) and an indication being given to the DTE that the 1200 bit/s channel is no longer ready (equivalent: turning OFF Cct 106).

Compliance shall be checked by the method described in Annex A subclause A.4.2.2

5.6.2.3 For modems which transmit carrier constantly when in the on-line state

The response times of the 1200 bit/s channel ready for sending indicator are the times that elapse between:

a) for the OFF to ON transition; an indication being given to the DTE that the modem is ready to operate (equivalent: turning ON Cct 107) and an indication being given to the DTE that it is able to transmit on the 1200 bit/s channel (equivalent: turning ON Cct 106);

b) for the ON to OFF transition; an indication being given to the DTE that the modem is no longer ready to operate (equivalent: turning OFF Cct 107) and an indication being given to the DTE that the 1200 bit/s channel is no longer ready (equivalent: turning OFF Cct 106).

The response times of the detector shall be within the limits stated in Table 3.

Compliance shall be checked by the method described in Annex A, subclause A.4.2.3

5.6.3 Response times of the backward channel ready for sending indicator

The applicant shall state whether the ready for sending delay is either a) or b) or both in Table 3, and that information shall be included in the instruction for use.

The response times of the backward channel ready for sending indicator are those given in subclauses 5.6.3.1 and 5.6.3.2.

5.6.3.1 For modems with DTE control

The response times of the backward channel ready for sending indicator are the times that elapse between:

a) for the OFF to ON transition: a DTE indicating that it wishes to transmit in the 75 bit/s channel (equivalent: turning ON Cct 120) and an indication being given to the DTE that the 75 bit/s channel is ready (equivalent: turning ON Cct 121);

b) for the ON to OFF transition: a DTE indicating that it no longer wishes to transmit in the 75 bit/s channel (equivalent: turning OFF Cct 120) and an indication being given to the DTE that the 75 bit/s channel is no longer ready (equivalent: turning OFF Cct 121).

Compliance shall be checked by the method described in Annex A subclause A.4.3.2.

5.6.3.2 For modems with control derived from the forward channel

For a modem in which control of the transmitted carrier in the 75 bit/s channel is controlled by the reception of signals from the remote modem in the 1200 bit/s channel, the response times of the backward channel ready for sending indicator are the times that elapse between:

a) for the OFF to ON transition: an indication to the DTE that a carrier signal has been received on the 1200 bit/s channel (equivalent: turning ON Cct 109) and an indication being given to the DTE that the 75 bit/s channel is ready (equivalent: turning ON Cct 121);

b) for the ON to OFF transition: an indication to the DTE that a carrier signal is no longer being received on the 1200 bit/s channel (equivalent: turning OFF Cct 109) and an indication being given to the DTE that the 75 bit/s channel is no longer ready (equivalent: turning OFF Cct 121).

The response times of the detector shall be within the limits stated in Table 3.

Compliance shall be checked by the method described in Annex A, subclause A.4.3.2.

Table 3: Ready for sending response times

Data Channel Ready for Sending Indicator	OFF to ON	750 ms to 1400 ms (Note 1)	a) 20 ms to 40 ms b) 200 ms to 275 ms
	ON TO OFF	(Note 2)	
Backward Chl. Signal Detector	OFF to ON	80 ms to 160 ms	
	ON TO OFF	(Note 2)	

NOTE 1: The longer range of response times is only to be used during call establishment in association with automatic calling and answering.

NOTE 2: The Ready for Sending ON to OFF delay has no effect upon the inter-operability between two modems. Therefore, no requirement exists.

5.7 Synchronous mode of use

Where the modem provides timing signals to facilitate synchronous transmission it shall generate a synchronization pattern to aid synchronization of the remote modem as follows.

A pattern of alternating binary 1 and binary 0 shall be transmitted throughout the period from the DTE, indicating that it wishes to send data (equivalent: turning ON Cct 105), and the modem indicating that it is ready to accept data from the DTE (equivalent: turning ON Cct 106).

Compliance shall be checked by the method described in Annex A, Clause A.5.

NOTE: Part of this synchronization pattern may appear as received data at the remote modem (equivalent: data on Cct 104) after this modem has indicated to the DTE that a carrier signal is being detected (equivalent: turning ON Cct 109). It will be necessary for a DTE to make provision for differentiating between these signals and genuine data signals.

5.8 Thresholds of received line signal detectors

The modem shall comply with the requirements of ETS 300 114 [2], subclause 5.4, when receiving a data signal of 1300 Hz for a modem capable of receiving at up to 1200 bit/s and 390 Hz for a modem capable of receiving at up to 75 bit/s.

5.9 Auto-calling and auto-answering procedures

5.9.1 Auto-calling - calling tone

This ETS does not require the provision of the calling tone defined in CCITT Recommendation V.25.

5.9.2 Auto-calling- recognition of answering tone

For asymmetrical duplex (i.e. modes b) and c) of subclause 5.2) modems to be used for automatically originated calls, the modem, or its associated call establishment equipment, shall comply with the requirements for answering tone detection given in ETS 300 114 [2], subclause 5.2.1.

5.9.3 Auto-answering - answer tone

For asymmetrical duplex (i.e. modes b) and c) of subclause 5.2) modems to be used for automatically answered calls, the modem, or its associated call answering equipment, shall comply with the requirements for answering tone generation given in ETS 300 114 [2], subclause 5.2.2.

5.10 Provision of test looping facilities

This ETS does not require the provision of the test looping facilities such as those defined in CCITT Recommendation V.54. However, the provision of such facilities does not preclude the approval to Category II.

5.11 Performance

NOTE : Since CCITT Recommendation V.23 [4] does not define performance criteria, the requirements of this clause are additional to the CCITT Recommendation.

5.11.1 Distortion

For an asynchronous modem, in each of the modes of operation for which approval as a Category II modem is sought, the following applies:

When two modems of the same type are interconnected via:

a) Test Line 3 (see ETS 300 114 [2], Annex C.5), the degree of synchronous start-stop distortion measured at the received data output (equivalent : Cct 104 for the forward channel, or 119 for the backward channel) of the modem under test, shall not be greater than 15% at 1200 bit/s in the 1200 bit/s channel and 15% at 75 bit/s in the 75 bit/s channel (if provided).

Compliance shall be checked by the method described in Annex A, subclause A.6.1.

b) Test Line 1, without frequency offset and the transient impairments, the degree of synchronous start-stop distortion measured at the received data output (equivalent : Cct 104 for the data channel, or 119 for the backward channel) of the modem under test shall not be greater than 35% at 1200 bit/s in the 1200 bit/s channel and 35% at 75 bit/s in the 75 bit/s channel (if provided).

Compliance shall be checked by the method described in Annex A, subclause A.6.2.

c) Test Line 1, without the transient impairments, but with a frequency shift of ± 4 Hz applied to the signals received by the modem under test, the degree of synchronous start-stop distortion measured at the received data output (equivalent : Cct 104 for data channel, or 119 for backward channel) of the modem under test shall not be greater than 35% at 1200 bit/s in the 1200 bit/s channel and 35% at 75 bit/s in the 75 bit/s channel (if provided).

Compliance shall be checked by the method described in Annex A, subclause A.6.3.

NOTE: Two modems of the "same type" is intended to convey that they are not only both V.23 but also from the same applicant/supplier. Where it is necessary to distinguish between the two modems, the one being approved is referred to as the "modem under test" and the other as the "modem used for reference".

When a non-reversible asymmetrical duplex modem (i.e. a modem handling only modes b) or c)) is to be tested, this shall be done against a modem used for reference. The modem used for reference may, on the choice of the applicant, be provided by the testing authority or by himself.

If serial binary data at the line signalling rate is not accessible at a digital interface of the modem under test, means shall be provided by the applicant to enable the testing authority to verify that data characters are correctly received by the modem under test.

5.11.2 Receiver performance

When tested as described in ETS 300 114 [2], subclause 5.6, the modem shall accumulate not less than 85 % error free seconds during each of the performance testing periods.

NOTE : The figure of 85% is provisional, and is based upon reasoned estimations.

Annex A (Normative): Testing methods

A.1 General testing conditions

The general conditions for test apply, as described in ETS 300 114 [2], Annex B.

For the testing of the modem it will generally be necessary to simulate the PSTN in the test set-up in order for the modem to remain in an on-line state. Unless otherwise indicated, Test Line 3 (see ETS 300 114 [2], Annex C, Clause C.5) should be used to connect the modem under test and the test apparatus.

A.2 Test for subclause 5.3.1 (Carrier frequencies)

The modem under test is caused to assume the on-line state and to transmit the frequencies associated with binary 1 and binary 0.

For those modems which are capable of transmitting at 75 bit/s,

- the frequency associated with binary 1, which shall be not less than 388 Hz and not greater than 392 Hz; and

- the frequency associated with binary 0, which shall be not less than 448 Hz and not greater than 452 Hz.

For those modems which are capable of transmitting at 1200 bit/s,

- the frequency associated with binary 1, which shall be not less than 1287 Hz and not greater than 1303 Hz; and

- the frequency associated with binary 0, which shall be not less than 2097 Hz and not greater than 2103 Hz.

A.3 Test for subclause 5.5 (Channel allocation)

The modem under test is connected to Test Line 3 (see ETS 300 114 [2] Annex C.5) used during test A.3.1. For the purpose of the tests in this clause a signal is present when the total signal power within the specified band exceeds a level of -30 dBm.

The modem under test is caused to enter the on-line state and the DTE to indicate that it no longer wishes to send data (equivalent: turning OFF Cct 105). The spectrum of the signal transmitted by the modem under test is checked to verify that signal power is present in the range 390 Hz to 450 Hz, but not in the range 1300 Hz to 2100 Hz. The DTE is now caused to indicate that it wishes to send data (equivalent: turning ON Cct 105). The spectrum of the signal transmitted by the modem under test is checked to verify that signal power is present in the range 1300 Hz to 2100 Hz.

A.4 Test for subclause 5.6 (Response times)

NOTE 1: The tolerance of ± 3,0 dB on the signal levels given below are to accommodate the differences between the insertion loss frequency response including the tolerances of the test line, and small differences in the transmit level.

NOTE 2: The tests for subclauses 5.6.1 and 5.6.2 could be performed together.

The modem under test is connected to Test Line 3 (see ETS 300 114 [2], Annex C.5) used during test A.3.1.

A.4.1 Test for subclause 5.6.1

1200 bit/s

For a modem which is capable of receiving at 1200 bit/s, the modem under test is caused to enter the on-line state, a tone is transmitted at 1300 Hz to arrive at the line terminals of the modem under test at a level of -20 dBm ± 3,0 dB. The time is measured from the moment of application of the signal until the modem indicates that a carrier signal has been detected.

The signal is then removed and the time measured between the instant of removal and the modem under test indicating that it is no longer detecting a carrier signal.

75 bit/s

For a modem which is capable of receiving at 75 bit/s, the modem under test is caused to enter the on-line state, a tone is transmitted at 390 Hz to arrive at the line terminals of the modem under test at a level of -20 dBm ± 3,0 dB. The time is measured from the moment of application of the signal until the modem under test indicates that a carrier signal has been detected.

The signal is then removed and the time measured between the instant of removal and the modem under test indicating that it is no longer detecting a carrier signal.

A.4.2 Test for subclause 5.6.2 (Data channel Ready For Sending (RFS) response times)

A.4.2.1 Test for subclause 5.6.2.1

For a modem which is capable of transmitting at 1200 bit/s, the modem under test is caused to enter the on-line state.

OFF to ON

The DTE is caused to indicate that it wishes to transmit (equivalent: turning ON Cct 105). The time is measured between the DTE indicating that it wishes to transmit and the modem under test indicating to the DTE that the 1200 bit/s data channel is ready (equivalent: turning ON Cct 106). The time measured shall be not less than 10 ms and not more than 20 ms except when a call is automatically established, when the time shall be not less than 300 ms and not more than 2000 ms following the first indication that the DTE wishes to transmit data (equivalent: turning ON Cct 105) occurring after an indication has been given to the DTE that the modem is ready to operate (equivalent: turning ON Cct 107).

A.4.2.2 Test for subclause 5.6.2.2

For a modem which is capable of transmitting at 1200 bit/s, the modem under test is caused to enter the on-line state.

OFF to ON

A tone is transmitted at 390 Hz to arrive at the line terminals of the modem under test at a level of -20 dBm ± 3,0 dB. The time is measured from the moment the modem under test indicates that a carrier signal has been detected (equivalent: turning ON Cct 122) and the modem under test indicating to the DTE that the 1200 bit/s data channel is ready (equivalent: turning ON Cct 106). The time measured shall be not less than 10 ms and not more than 20 ms except when a call is automatically established, when the time shall be not less than 300 ms and not more than 2000 ms following the first indication that the DTE wishes to transmit data (equivalent: turning ON Cct 105) occurring after an indication has been given to the DTE that the modem is ready to operate (equivalent: turning ON Cct 107).

A.4.2.3 Test for subclause 5.6.2.3

For a modem which is capable of transmitting at 1200 bit/s, the modem under test is caused to enter the on-line state.

OFF to ON

The DTE is caused to indicate that it wishes the modem to assume an on-line state (equivalent: turning ON Cct 108), or the modem is caused to assume the on-line state by the method indicated by the applicant. The time is measured between the modem under test indicating that it is ready to operate (equivalent: turning ON Cct 107) and the modem under test indicating to the DTE that the 1200 bit/s data channel is ready (equivalent: turning ON Cct 106). The time measured shall be not less than 10 ms and not more than 20 ms except when a call is automatically established, when the time shall be not less than 300 ms and not more than 2000 ms following the first indication that the DTE wishes to transmit data (equivalent: turning ON Cct 105) occurring after an indication has been given to the DTE that the modem is ready to operate (equivalent: turning ON Cct 107).

A.4.3 Test for subclause 5.6.3 (Backward channel RFS response times)

A.4.3.1 Test for subclause 5.6.3.1

For a modem which is capable of transmitting at 75 bit/s, the modem under test is caused to enter the on-line state.

OFF to ON

The DTE is caused to indicate that it wishes to transmit (equivalent: turning ON Cct 120). The time is measured between the DTE indicating that it wishes to transmit and the modem under test indicating to the DTE that the 75 bit/s data channel is ready (equivalent: turning ON Cct 121). The time measured shall be not less than 80 ms.

A.4.3.2 Test for subclause 5.6.3.2

For a modem which is capable of transmitting at 75 bit/s, the modem under test is caused to enter the on-line state.

OFF to ON

A tone is transmitted at 390 Hz to arrive at the line terminals of the modem under test at a level of -20 dBm ± 3,0 dB. The time is measured from the moment the modem under test indicates that a carrier signal has been detected (equivalent: turning ON Cct 122) and the modem under test indicating to the DTE that the 75 bit/s data channel is ready (equivalent: turning ON Cct 121). The time measured shall be not less than 80 ms.

A.5 Test for subclause 5.7 (Synchronous mode of use)

For a modem which is capable of transmitting at 1200 bit/s in the synchronous mode of use, the modem under test is connected to the modem used for reference via Test Line 3 (see ETS 300 114 [2], Annex C.5). After the modem used for reference having given an indication that it has detected a line signal in the forward channel(equivalent: turning ON Cct 109), an indication shall be given from the DTE connected to the modem under test that it wishes to transmit data. During the time until the modem under test indicates to the DTE that it is ready to accept data from the DTE (equivalent: turning ON Cct 106), alternating binary 1 and binary 0 shall be output at the received data lead of the modem used for reference (equivalent: Cct 104).

A.6 Test for subclause 5.11.1 (Distortion)

The following test pattern is specified for the tests described in subclauses A.6.1, A.6.2 and A.6.3:

A cyclic string of discrete test characters (DTCs) consisting of:

DTC N° 2 + DTC N° 3 + DTC N° 2 + DTC N° 1 + DTC N° 2 + ...

as specified in ETS 300 114 [2], Annex B, subclause B.5.1.3 for the 10 bit character length.

The test period for the tests described in subclauses A.6.1, A.6.2 and A.6.3 shall be one minute.

A.6.1 Test for subclause 5.11.1 (a)

The modem under test is connected to the modem used for reference via Test Line 3 (see ETS 300 114 [2], Annex C.5). The modem used for reference is caused to transmit the test pattern to the modem under test at rate equal to the maximum bit rate for the channel under test i.e. 1200 bit/s or 75 bit/s. The degree of synchronous start-stop distortion of the received data output (equivalent : Cct 104 for the data channel, or Cct 119 for backward channel) of the modem under test shall not be greater than 15%.

A.6.2 Test for subclause 5.11.1 (b)

The modem under test is connected to the modem used for reference via Test Line 1 without frequency offset (see ETS 300 114 [2], Annexes B.6.4 and C.3) and transient impairments. The modem used for reference is caused to transmit the test pattern to the modem under test at rate equal to the maximum bit rate for the channel under test i.e. 1200 bit/s or 75 bit/s. The degree of synchronous start-stop distortion of the received data output (equivalent: Cct 104 for the data channel, or Cct 119 for backward channel) of the modem under test shall not be greater than 35%.

A.6.3 Test for subclause 5.11.1 (c)

This test shall be performed twice once with a frequency offset of +4 Hz and then with a frequency offset of -4 Hz.

The modem under test is connected to the modem used for reference via Test Line 1 without transient impairments (see ETS 300 114 [2], Annexes B.6.4 and C.3). The modem used for reference is caused to transmit the test pattern to the modem under test at rate equal to the maximum bit rate for the channel under test i.e. 1200 bit/s or 75 bit/s. The degree of synchronous start-stop distortion of the received data output (equivalent: Cct 104 for the data channel, or Cct 119 for backward channel) of the modem under test shall not be greater than 35%.

Annex B (Informative): Proforma declaration of modes of operation/use

Declaration of modes of operation/use for 1200 bit/s half duplex and 1200/75 bit/s asymmetrical duplex modems based on CCITT Recommendation V.23 [4]

The modem submitted for approval is capable of operating in the modes indicated in the table below.

Table B1: Proforma for declaring modes of use

1200 bit/s half duplex	
a) Asynchronous	
d) Synchronous	
asymmetrical duplex	
b) 1200/75 bit/s asymmetrical duplex	
c) 75/1200 bit/s asymmetrical duplex	

Table B2: Proforma for declaring modes of operation/use

Auto calling or answering	
a) Auto calling	
b) Auto answering	

Table B3: Proforma for declaring modes of operation/use

1200 bit/s channel ready for sending indicator	
a) DTE control	
b) Control derived from the 1200 bit/s channel	
c) neither of the above methods	

Table B4 Proforma for declaring modes of operation/use

75 bit/s channel ready for sending indicator	
a) DTE control	
b) Control derived from the 75 bit/s channel	

Table B5: Proforma for declaring modes of operation/use

Ready for sending response times	
a) 750 ms to 1400 ms for initial attempt	
b) 20 ms to 40 ms	
c) 200 ms to 275 ms	

Table B6: Proforma for declaring modes of operation/use

Channel allocation for asymmetrical duplex modems	
a) Transmit on backward channel when forward channel is off	
b) Separate control of the forward and backward channels by the DTE	

History

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